

Cloud Operations

1. Café business Client requires a solution for hosting static website as they wish to promote café products to their customers.



Task 1: Connect to an Amazon Linux EC2 instance by using SSH

MacOS/Linux – Connect Using SSH

- Choose the Details drop down and then choose Show a credentials window will open
- Choose the Download PEM button and save the labsuser.pem file.
- Open a terminal window and change directory `cd` to the directory where the labsuser.pem file was downloaded.
`cd ~/Downloads`
- Change the permissions on the key.
`chmod 600 labsuser.pem`
- Go to AWS Management Console, and in the EC2 service, choose Instances, Select the instances that you want to connect to.
- Copy the IPV4 public IP value
- Return to the terminal window and connect to instance via SSH
`ssh -i labsuser.pem ec2-user@<public-ip>`
replace **<public-ip>** with the actual public IP address you copied).

Task 2: Configure the AWS CLI

- Update the AWS CLI software with the credentials

aws configure

- At the prompts enter the following information
 - I. AWS Access key ID: Select the Details drop down menu and choose Show. Copy the AccessKey value.
 - II. AWS Secret Access Key: Copy and paste the SecretKey value from the same Credentials screen.
 - III. Default region name: us-east-1
 - IV. Default output format: json

Task 3: Create an S3 bucket by using the AWS CLI

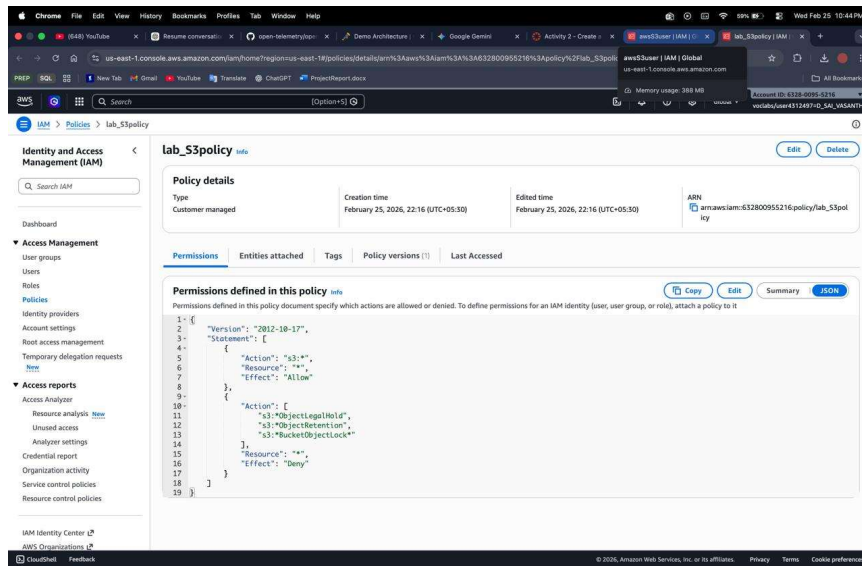
- Use the aws s3api create-bucket command to create a bucket in Amazon S3.
- Choose a Unique Bucket Name
- Specify region because you want to host the website in the AWS Region

```
aws s3api create-bucket \  
--bucket <your-unique-bucket-name> \  
--region us-east-1 \  
--create-bucket-configuration LocationConstraint=us-east-1
```

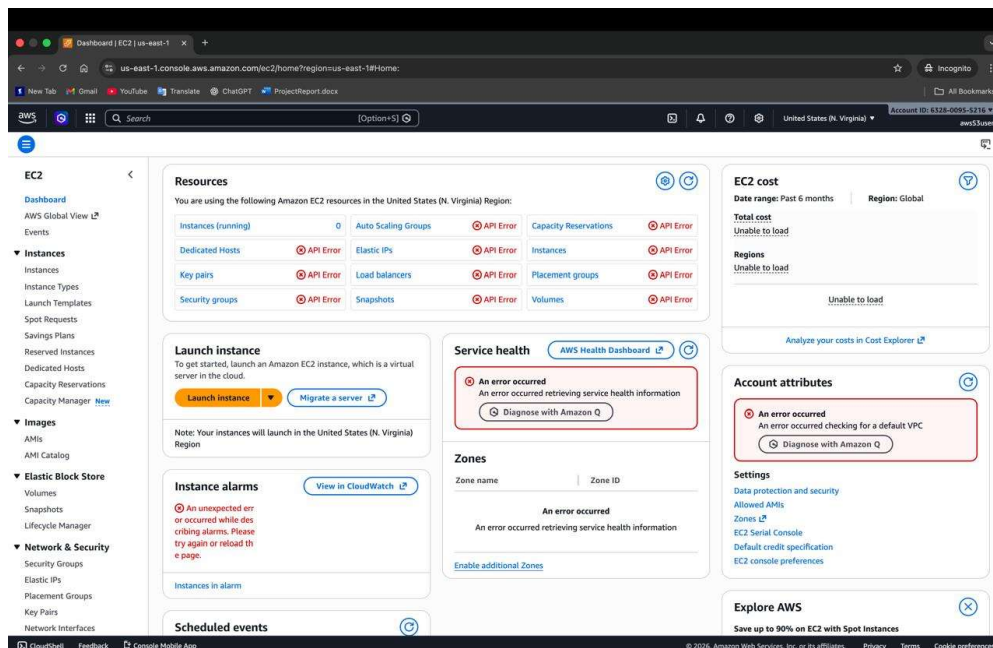
- If the command is successful, you will get a JSON-formatted response.

Task 4: Analyse details of an IAM user that has full access to Amazon S3

- In the AWS Management Console, Type IAM and choose IAM service.
- In the left Navigation pane, choose users and then choose awsS3user.
- Choose the permissions tab, expand the policy named lab_S3policy and click {} JSON to view the policy details.
- The policy contains two statements
 - I. Allows full access to Amazon S3
 - II. Uses wildcard s3: *, meaning all S3 actions are allowed
- The second statements deny some specific S3 object-related actions
- Choose the Security credentials tab, Notice an existing Access Key
- In the Security Credentials tab, Copy the Console sign-in link



- Open a Private Browser Window
- Paste the Console sign-in URL into the new browser window and sign in as IAM user.
- In the console choose Services, select Amazon EC2
- You will see multiple API Error message, this is expected because IAM user does not have EC2 permissions.
- Choose Services, Select Amazon S3
- S3 console loads without errors, you can see the bucket you created earlier, This confirms the IAM policy grants S3 access.



Task 5: Extract the files that you need for this activity

- In the SSH terminal, extract the files
`cd ~/sysops-activity-files tar xvzf static-website.tar.gz`
- Run the `ls` command, you will see an `index.html` file and directories named `css` and `images`.
- Delete the `tar.gz` file using `rm` command but leave the extracted files.

```
[ec2-user@ip-10-200-0-55 ~]$ aws s3 ls
2026-02-25 17:00:01 vassu12
[ec2-user@ip-10-200-0-55 ~]$ ls
sysops-activity-files
[ec2-user@ip-10-200-0-55 ~]$ cd sysops-activity-files/
[ec2-user@ip-10-200-0-55 sysops-activity-files]$ tar xvzf static-website.tar.gz
css/
css/styles.css
images/
images/Cake-Vitrine.png
images/Coffee-and-Pastries.png
images/Cookies.png
images/Cup-of-Hot-Chocolate.png
images/Mom-&-Pop-Coffee-Shop.png
images/Mom-&-Pop.png
images/Strawberry-&-Blueberry-Tarts.png
images/Strawberry-Tarts.png
index.html
[ec2-user@ip-10-200-0-55 sysops-activity-files]$
```

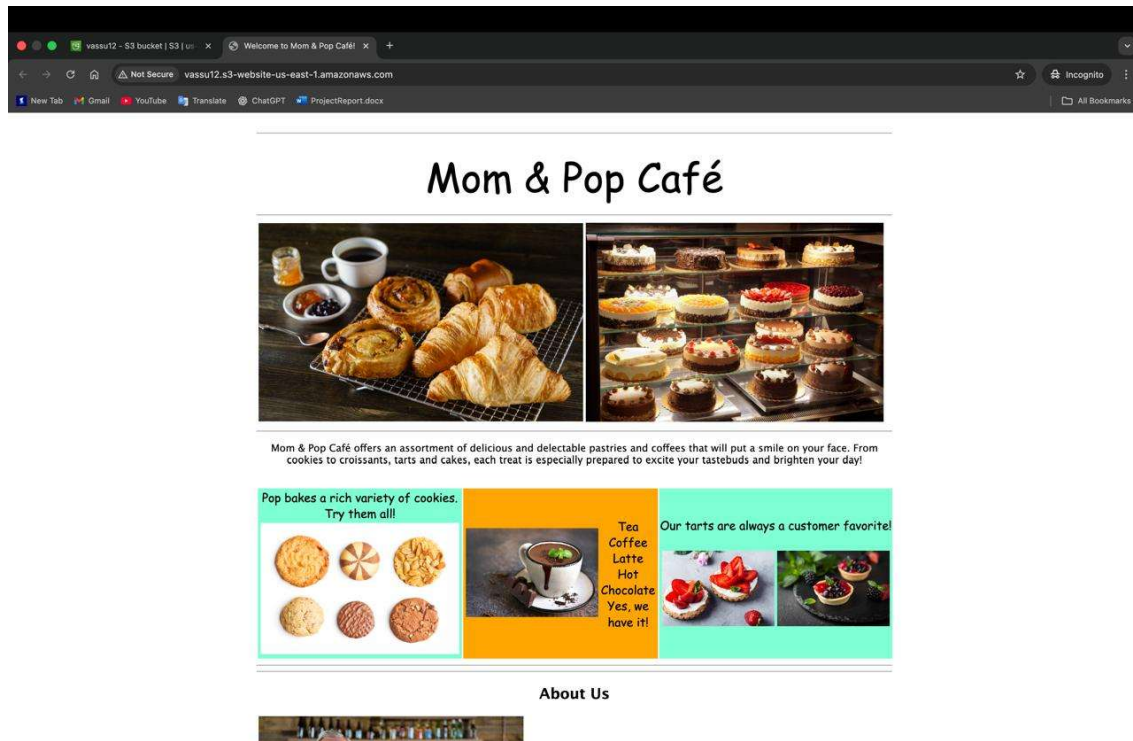
Task 6: Update bucket controls/Upload files to Amazon S3 by using the AWS CLI

- Configure Object Ownership
`aws s3api put-bucket-ownership-controls \`
`--bucket <my-bucket> \`
`--ownership-controls "Rules=[{ObjectOwnership=BucketOwnerPreferred}]"`
- Disable Public Access Restrictions
`aws s3api put-public-access-block \`
`--bucket <my-bucket> \`
`--public-access-block-configuration`
`"BlockPublicAcls=false,IgnorePublicAcls=false,BlockPublicPolicy=false,RestrictPublicBuckets=false"`
- Configure Website Hosting
`aws s3 website s3://<my-bucket>/ --index-document index.html`
- Upload files to the Bucket

```
aws s3 cp . s3://<my-bucket> --recursive --acl public-read
```

- Verify Upload: `aws s3 ls <my-bucket>`
- Verify Website in AWS Console
 - I. Go to the Amazon Web Services Management Console
 - II. Open Amazon s3
 - III. Select your bucket
 - IV. Choose the Properties tab
 - V. Scroll to the Static website hosting section
 - VI. Confirm that hosting is Enabled
 - VII. Click the Bucket website endpoint link

```
[ec2-user@ip-10-200-0-55 ~]$ aws s3 ls
2024-02-25 17:00:01 vassul2
[ec2-user@ip-10-200-0-55 ~]$ ls
[ec2-user@ip-10-200-0-55 ~]$ cd s3ops-activity-files/
[ec2-user@ip-10-200-0-55 s3ops-activity-files]$ tar xvf static-website.tar.gz
css/
css/styles.css
images/
images/Cake-Vitrine.png
images/Coffee-and-Pastries.png
images/Cookies.png
images/Cup-of-Hot-Chocolate.png
images/Mom-A-Pop-Coffee-Shop.png
images/Mom-A-Pop.png
images/Strawberry-A-Blueberry-Tarts.png
images/Strawberry-Tarts.png
index.html
[ec2-user@ip-10-200-0-55 s3ops-activity-files]$ aws s3api put-bucket-ownership-controls \
> --bucket <my-bucket> \
> --tar xvf static-website.tar.gz
[ec2-user@ip-10-200-0-55 s3ops-activity-files]$ aws s3api put-bucket-ownership-controls --bucket <my-bucket> --ownership-controls "Rules=[{ObjectOwnership=BucketOwnerPreferred}]"
-bash: my-bucket: No such file or directory
[ec2-user@ip-10-200-0-55 s3ops-activity-files]$ aws s3api put-bucket-ownership-controls --bucket vassul2 --ownership-controls "Rules=[{ObjectOwnership=BucketOwnerPreferred}]"
[ec2-user@ip-10-200-0-55 s3ops-activity-files]$ aws s3api put-public-access-block --bucket vassul2 --public-access-block-configuration "BlockPublicAcl=false,IgnorePublicAcls=false,BlockPublicPolicy=false,RestrictPublicBuckets=false"
[ec2-user@ip-10-200-0-55 s3ops-activity-files]$ aws s3 website s3://vassul2 --index-document index.html
[ec2-user@ip-10-200-0-55 s3ops-activity-files]$ aws s3 cp . s3://vassul2/ --recursive --acl-public-read
Unknown options: --acl-public-read
[ec2-user@ip-10-200-0-55 s3ops-activity-files]$ aws s3 cp . s3://vassul2/ --recursive --acl public-read
upload: css/styles.css to s3://vassul2/css/styles.css
upload: ./index.html to s3://vassul2/index.html
upload: images/Mom-A-Pop-Coffee-Shop.png to s3://vassul2/images/Mom-A-Pop-Coffee-Shop.png
upload: images/Coffee-and-Pastries.png to s3://vassul2/images/Coffee-and-Pastries.png
upload: images/Cake-Vitrine.png to s3://vassul2/images/Cake-Vitrine.png
upload: images/Strawberry-A-Blueberry-Tarts.png to s3://vassul2/images/Strawberry-A-Blueberry-Tarts.png
upload: images/Cookies.png to s3://vassul2/images/Cookies.png
upload: images/Mom-A-Pop.png to s3://vassul2/images/Mom-A-Pop.png
upload: images/Strawberry-Tarts.png to s3://vassul2/images/Strawberry-Tarts.png
upload: images/Cup-of-Hot-Chocolate.png to s3://vassul2/images/Cup-of-Hot-Chocolate.png
upload: ./static-website.tar.gz to s3://vassul2/static-website.tar.gz
[ec2-user@ip-10-200-0-55 s3ops-activity-files]$ aws s3 ls
2024-02-25 17:00:01 vassul2
[ec2-user@ip-10-200-0-55 s3ops-activity-files]$ ls
css images index.html static-website.tar.gz
[ec2-user@ip-10-200-0-55 s3ops-activity-files]$ cd -
~
[ec2-user@ip-10-200-0-55 s3ops-activity-files]$ cd -
~
[ec2-user@ip-10-200-0-55 s3ops-activity-files]$ cd -
[ec2-user@ip-10-200-0-55 ~]$ touch update-website.sh
[ec2-user@ip-10-200-0-55 ~]$ nano update-website.sh
[ec2-user@ip-10-200-0-55 ~]$
```



Task 7: Create a batch file to make updating the website easily repeatable

- Navigate to Home Directory: `cd`
- Create an Empty Script File: `touch update-website.sh`
Open the File in nano editor: `Nano update-website.sh`
- Add Script Content
`#!/bin/bash`
`Aws s3 cp ~/sysops-activity-files/s3://<my-bucket>/ --recursive --acl public read`
- Make the Script Executable: `chmod +x update-website.sh`
Open the index.html: `nano sysops-activity-files/index.html`
- Change the first `bgcolor="aquamarine"` to `bgcolor="gainsboro"`
- Change the `bgcolor="orange"` to `bgcolor="aquamarine"`
- Locate the second line that has the HTML code `bgcolor="aquamarine"` and change it to `bgcolor="gainsboro"`
- Run the batch file: `./update-website.sh`
- Open browser and refresh the mom and pop café webpage to see the changes to the website.

```

[ec2-user@ip-10-200-0-55 ~]$ nano sysops-activity-files/index.html
[ec2-user@ip-10-200-0-55 ~]$ ./update-website.sh
-bash: ./update-website.sh: bin/bash: bad interpreter: No such file or directory
[ec2-user@ip-10-200-0-55 ~]$ nano update-website.sh
[ec2-user@ip-10-200-0-55 ~]$ chmod +x update-website.sh
[ec2-user@ip-10-200-0-55 ~]$ ./update-website.sh
upload: .aws/config to s3://vassu12/.aws/config
upload: .aws/credentials to s3://vassu12/.aws/credentials
upload: ./bash_history to s3://vassu12/.bash_history
upload: ./bash_input to s3://vassu12/.bash_input
upload: ./bash_profile to s3://vassu12/.bash_profile
upload: .ssh/authorized_keys to s3://vassu12/.ssh/authorized_keys
upload: ./awserr to s3://vassu12/.awserr
upload: sysops-activity-files/css/styles.css to s3://vassu12/sysops-activity-files/css/styles.css
upload: ./update-website.sh to s3://vassu12/update-website.sh
upload: sysops-activity-files/index.html to s3://vassu12/sysops-activity-files/index.html
upload: sysops-activity-files/images/Cookies.png to s3://vassu12/sysops-activity-files/images/Cookies.png
upload: sysops-activity-files/images/Mom-4-Pop-Coffee-Shop.png to s3://vassu12/sysops-activity-files/images/Mom-4-Pop-Coffee-Shop.png
upload: sysops-activity-files/images/Mom-4-Pop.png to s3://vassu12/sysops-activity-files/images/Mom-4-Pop.png
upload: sysops-activity-files/images/Cup-of-Hot-Chocolate.png to s3://vassu12/sysops-activity-files/images/Cup-of-Hot-Chocolate.png
upload: sysops-activity-files/images/Coffee-and-Pastries.png to s3://vassu12/sysops-activity-files/images/Coffee-and-Pastries.png
upload: sysops-activity-files/images/Cake-Vintage.png to s3://vassu12/sysops-activity-files/images/Cake-Vintage.png
upload: sysops-activity-files/images/Strawberry-4-Blueberry-Tarts.png to s3://vassu12/sysops-activity-files/images/Strawberry-4-Blueberry-Tarts.png
upload: sysops-activity-files/images/Strawberry-Tarts.png to s3://vassu12/sysops-activity-files/images/Strawberry-Tarts.png
upload: sysops-activity-files/static-website.tar.gz to s3://vassu12/sysops-activity-files/static-website.tar.gz
[ec2-user@ip-10-200-0-55 ~]$ nano sysops-activity-files/index.html
[ec2-user@ip-10-200-0-55 ~]$ nano update-website.sh
[ec2-user@ip-10-200-0-55 ~]$ ./update-website.sh

Unknown options: --recursive
[ec2-user@ip-10-200-0-55 ~]$ nano update-website.sh
[ec2-user@ip-10-200-0-55 ~]$ ./update-website.sh
upload: sysops-activity-files/index.html to s3://vassu12/sysops-activity-files/index.html
upload: ./update-website.sh to s3://vassu12/update-website.sh
[ec2-user@ip-10-200-0-55 ~]$

```

2. Café business Client requires a solution for hosting dynamic website as their users can pre-order café products. As a cloud operation engineer provide a solution for implementing the business needs.

Task 1: Connect to an Amazon Linux EC2 instance by using SSH

MacOS/Linux – Connect Using SSH

- Choose the Details drop down and then choose Show a credentials window will open
- Choose the Download PEM button and save the labsuser.pem file.
- Open a terminal window and change directory cd to the directory where the labsuser.pem file was downloaded.

```
cd ~/Downloads
```
- Change the permissions on the key.

```
chmod 600 labsuser.pem
```
- Go to AWS Management Console, and in the EC2 service, choose Instances, Select the instances that you want to connect to.
- Copy the IPV4 public IP value
- Return to the terminal window and connect to instance via SSH

```
ssh -i labsuser.pem ec2-user@<public-ip>
```

replace **<public-ip>** with the actual public IP address you copied).

Task 2: Configure the AWS CLI

- Update the AWS CLI software with the credentials
aws configure
- At the prompts enter the following information
 - V. AWS Access key ID: Select the Details drop down menu and choose Show. Copy the AccessKey value.
 - VI. AWS Secret Access Key: Copy and paste the SecretKey value from the same Credentials screen.
 - VII. Default region name: us-east-1
 - VIII. Default output format: json

Task 3: Create an EC2 instance by using the AWS CLI

- Navigate to the Script Directory: `cd ~/sysops-activity-files/starters`
- Create a Backup of the Script: `cp create-lamp-instance.sh create-lamp-instance.backup`
- Open the create-lamp-instance.sh in text editor: `nano create-lamp-instance.sh`
- Analyze the contents of the script
 - i. Line1- `#!/bin/bash` This is a bash file
 - ii. Lines 7-11: The instance size is set to `t2.small`
 - iii. Lines 16-29: The script invokes the AWS CLI `describe-regions` command to get a list of all AWS regions. It searches each region for a VPC named MomPopCafe VPC. When it finds it captures the vpc ID, captures the region and exits the loop.
 - iv. Lines 31-57: The script invokes AWS CLI commands to look up the subnet ID, Keypair name, and AMI ID values that will be needed to create an instance.
 - v. Lines 59-124: Cleanup section, before creating resources, the script checks for existing ones like EC2 instance named mompopcafe server and SG named mompopcafe SG. If found prompts to delete them.

- vi. Lines 126-154: The script creates a new security group with ports 22 and 80 open.
 - vii. Lines 156-168: The script creates a new EC2 instance.
 - viii. Lines 179-188: Extracts the instance Id from the instance details, runs a loop 10 seconds for a public IP. Once assigned displays the public IP address
- View the user-Data Script: `cat create-lamp-instance-userdata.txt`
 - Run the script: `./create-lamp-instance.sh`

Task 4: Troubleshoot issues

- Issue 1: The terminal output states an error `InvalidSubnetID.NotFound`
- Fix: Verify resources like AMI ID or subnet exist in every region the rerun the script.
- Issue 2: SSH Permission Denied
- Fix: Ensure the correct username, ensure if file permissions on the key pair file set correctly. Reconnect to the instance.
- Issue 3: Website not loading
- Fix:
 - i. verify web server running - `sudo systemctl status httpd`
 - ii. If not running, then – `sudo systemctl start httpd`
 - iii. Install Nmap which is port scanning tool– `sudo yum install -y nmap`
 - iv. To scan ports – `nmap -Pn <public-ip>`
 - v. The results that are returned by this command show which ports are accessible – 22/tcp open, 80/tcp open.
- Verify webpage: open browser and type `http://<public-ip>`
- Expected message: Hello from your web server
- Verify User-Data Script Execution: `sudo tail -f /var/log/cloud-init-output.log`

```
[ec2-user@cli-host starters]$ ./create-lamp-instance.sh
Running create-instance.sh on 2020-02-26 05:29:57
Region: us-east-1
Instance Type: t2.small
Profile: default

Looking up account values...
VPC: vpc-9f4a79a5b3ba0a0f
Subnet ID: subnet-e8c0ff93dc22999ba
Key: kockey
AMI ID: ami-8199fa5fada510433

Creating a new security group...
Security Group: sg-81f29c4e22758329a

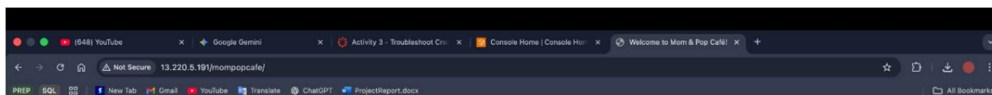
Opening port 22 in the new security group
Opening port 80 in the new security group
Creating an EC2 instance in us-east-1

Instance Details...
{
  "Groups": [{}],
  "Instances": [
    {
      "AmiLaunchIndex": 0,
      "Architecture": "x86_64",
      "BlockDeviceMappings": [{}],
      "CapacityReservationSpecification": {
        "CapacityReservationPreference": "open"
      },
      "ClientToken": "8dc87779-1c17-4ce1-9ba3-8ec13b2138d3",
      "CpuOptions": {
        "CoreCount": 1,
        "ThreadsPerCore": 1
      },
      "EbsOptimized": false,
      "EnaSupport": true,
      "Hypervisor": "xen",
      "ImageId": "ami-8199fa5fada510433",
      "InstanceId": "i-88a7c7a8b70bba07",
      "InstanceType": "t2.small",
      "LaunchTime": "2020-02-26T05:30:06.000Z",
      "MetadataOptions": {
        "HttpEndpoint": "enabled",
        "HttpPutResponseHopLimit": 1,
        "HttpTokens": "optional",
        "State": "pending"
      },
      "Monitoring": {
        "State": "disabled"
      },
      "NetworkInterface": [
        {
          "Attachment": {
            "AttachTime": "2020-02-26T05:30:06.000Z",
            "AttachmentId": "eni-attach-8da720ba70bba07",
            "DeleteOnTermination": true,
            "DeviceIndex": 0,
            "Status": "attaching"
          },
          "Description": "",
          "Groups": [
            {
              "GroupId": "sg-81f29c4e22758329a",
              "GroupName": "mompopcafeSG"
            }
          ],
          "InterfaceType": "interface",
          "Ipv6Addresses": [{}],
          "MacAddress": "82:78:fa:02:3e:1d",
          "NetworkInterfaceId": "eni-8588013a7898a9935",

```

Task 5: Verify the functionality of the new website

- Load `http://<public-ip>/mompopcafe` in the web browser, the improved Mompopcafe website loads successfully
- Test the Menu Page: select quantities for dessert items and submit an order. The order confirmation page displays an items list.



About Us

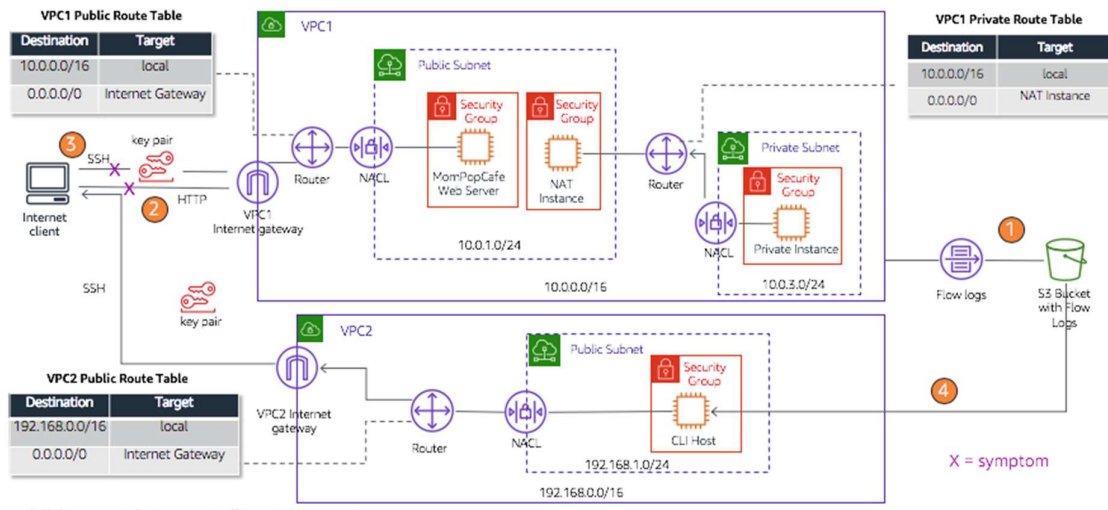
Mom & Pop Café

Home Menu Order History

Pastries

 Croissant \$1.50 Fresh, buttery and fluffy... Simply delicious! Quantity: 0	 Donut \$1.00 We have more than half-a-dozen flavors! Quantity: 0	 Chocolate Chip Cookie \$2.50 Made with Swiss chocolate with a touch of Madagascar vanilla Quantity: 0
 Muffin \$3.00 Banana bread, blueberry, cranberry or apple Quantity: 0	 Strawberry Blueberry Tart \$3.50 Bursting with the taste and aroma of fresh fruit Quantity: 0	 Strawberry Tart \$3.50 Made with fresh ripe strawberries and a delicious whipped cream Quantity: 0

3. Café business client wishes to have an isolated networking space for provisioning services and infrastructure for his banking dynamic application. Client also wishes to deploy front-end of his application open to public and needs a private and secure environment for his database layer. Based on the requirements, VPC is created with public subnets hosting web server and private subnets with database servers.



Task 1: Connect to the CLI Host EC2 instance by using SSH

MacOS/Linux – Connect Using SSH

- Choose the Details drop down and then choose Show a credentials window will open
- Choose the Download PEM button and save the labsuser.pem file.
- Open a terminal window and change directory `cd` to the directory where the labsuser.pem file was downloaded.
`cd ~/Downloads`
- Change the permissions on the key.
`chmod 600 labsuser.pem`
- Go to lab instructions, click Details, select Show from the dropdown menu, locate the field labeled CliHostIP and copy the IPV4 Public IP address.
- Return to the terminal window and connect to instance via SSH
`ssh -i labsuser.pem ec2-user@<public-ip>`
 replace **<public-ip>** with the actual public IP address you copied).

Task 2: Configure the AWS CLI

- Discover the region in which the CLI Host instance is running
`curl http://169.254.169.254/latest/dynamic/instance-identity/document | grep region`
- Update the AWS CLI software with the credentials
`aws configure`
- At the prompts enter the following information
 - IX. AWS Access key ID: Select the Details drop down menu and choose Show.
Copy the AccessKey value.
 - X. AWS Secret Access Key: Copy and paste the SecretKey value from the same Credentials screen.
 - XI. Default region name: us-east-1
 - XII. Default output format: json

Task 3: Enable VPC Flow Logs

- Create the S3 bucket that will hold the flow logs
`aws s3api create-bucket --bucket flowlog#### --region<region> --create-bucket-configuration LocationConstraint=<region>`

replace #### with four random numbers
<region> with the region where your EC2 instances were created
- Get VPC ID for VPC1
`aws ec2 describe-vpcs --query 'Vpcs[*].[VpcId,Tags[?Key==`Name`].Value,CidrBlock]' --filters "Name=tag:Name,Values='VPC1'"`
- Enable VPC Flow logs on VPC1

`aws ec2 create-flow-logs --resource-type VPC --resource-ids <vpc-id> --traffic-type ALL --log-destination-type s3 --log-destination arn:aws:s3:::<flowlog####>`

Replace <vpc-id> with the actual VPC ID of VPC1
<flowlog####> with your actual S3 bucket name
- If the command runs successfully, you should see that a FlowLogId and ClientToken are returned
- Confirm the Flow Log was created
`aws ec2 describe-flow-logs`

- Verify that output should show a single flow log was created with a FlowLogStatus of ACTIVE and log destination points to S3 bucket.

```
[ec2-user@cli-host ~]$ aws s3api create-bucket --bucket flowlogs --region us-east-1
An error occurred (BucketAlreadyExists) when calling the CreateBucket operation: The requested bucket name is not available. The bucket namespace is shared by all users of the system. Please select a different name and try again.
[ec2-user@cli-host ~]$ aws s3api create-bucket --bucket flowlogs26 --region us-east-1
{
  "Location": "/flowlogs26"
}
[ec2-user@cli-host ~]$ aws ec2 describe-vpcs --query 'Vpcs[0].[VpcId,Tags[?Key==`Name`].Value,CidrBlock]' --filters 'Name=tag:Name,Values=vpc1'
{
  "Vpcs": [
    {
      "VpcId": "vpc-0e402a13823b6cf5b",
      "VPC1": "10.0.0.0/16"
    }
  ]
}
[ec2-user@cli-host ~]$ aws ec2 create-flow-logs --resource-type VPC --resource-ids <vpc-id> --traffic-type ALL --log-destination-type s3 --log-destination arn:aws:s3:::flowlogs26
-bash: vpc-id: No such file or directory
[ec2-user@cli-host ~]$ aws ec2 create-flow-logs --resource-type VPC --resource-ids <vpc-id> --traffic-type ALL --log-destination-type s3 --log-destination arn:aws:s3:::flowlogs26
-bash: vpc-id: No such file or directory
[ec2-user@cli-host ~]$ aws ec2 create-flow-logs --resource-type VPC --resource-ids vpc-0e402a13823b6cf5b --traffic-type ALL --log-destination-type s3 --log-destination arn:aws:s3:::flowlogs26
{
  "Unsuccessful": [],
  "FlowLogIds": [
    "fl-085a3a3d6f63ad55"
  ],
  "ClientToken": "zFlmpFv8kpa3YtZp5WvLzNzW69tCWdMzFh3Gcn"
}
[ec2-user@cli-host ~]$ aws ec2 describe-flow-logs
{
  "FlowLogs": [
    {
      "LogDestinationType": "s3",
      "Tags": [],
      "ResourceId": "vpc-0e402a13823b6cf5b",
      "CreationTime": "2024-02-26T06:15:01.657Z",
      "TrafficType": "ALL",
      "FlowLogStatus": "ACTIVE",
      "LogFormat": "${version} ${account-id} ${interface-id} ${srcaddr} ${dstaddr} ${srcport} ${dstport} ${protocol} ${packets} ${bytes} ${start} ${end} ${action} ${log-status}",
      "FlowLogId": "fl-085a3a3d6f63ad55",
      "MaxAggregationInterval": 600,
      "LogDestination": "arn:aws:s3:::flowlogs26",
      "DeliverLogsStatus": "SUCCESS"
    }
  ]
}
```

Task 4: Analyze and troubleshoot access to resources

- Copy the Web Server Public IP address from the lab instructions
- Test web Access – open a new browser tab and paste the WebServerIP, the page fails to load with ERR_CONNECTION_TIMED_OUT
- In the terminal verify instance details


```
aws ec2 describe-instances --filter "Name=ip-address,Values='<WebServerIP>'"
```
- Filter relevant details like State of the instance, the private IP address, the instance ID, the security group, subnet ID and key pair name.

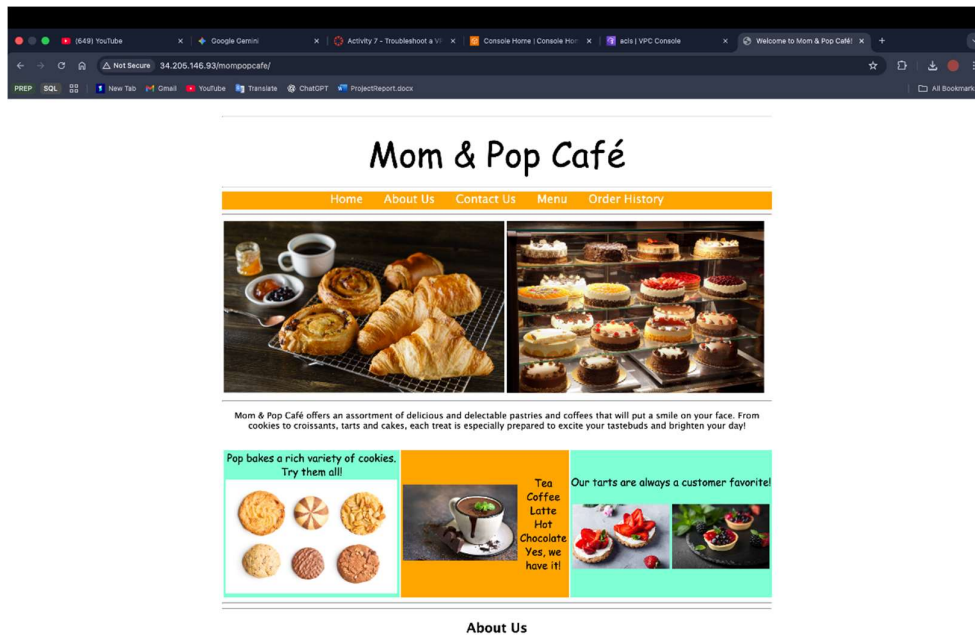

```
aws ec2 describe-instances --filter "Name=ip-address,Values='<WebServerIP>'" --query 'Reservations[*].Instances[*].[State,PrivateIpAddress,InstanceId,SecurityGroups,SubnetId,KeyName]'
```
- To test SSH connectivity disconnect from CLI host by entering exit
- macOS and Linux users


```
ssh -i labsuser.pem ec2-user@<WebServerIP> -o ConnectTimeout=10
```
- Try establishing SSH connection with 15-second timeout, it fails again.
- Re-establish the SSH connection to the CLI Host instance


```
ssh -i labsuser.pem ec2-user@CliHostIP
```

- Troubleshooting Challenge 1: Web Access Issue
 - i. Problem: Instance is running However cannot load the web page and also cannot use SSH to connect to the web server instance.
 - ii. Hint:
 - A. Check open ports using nmap. To do this first install the utility on the CLI Host - sudo yum install -y nmap then run nmap <WebServerIP> if no ports are open, something is blocking traffic.
 - B. Check the security group by running command aws ec2 describe-security-groups Verify inbound rules like HTTP (port 80) and SSH (port 22).
 - C. Check Route Table
 - aws ec2 describe-route-tables --filter "Name=association.subnet-id,Values='<VPC1PubSubnetID>'",
 - verify public subnet has route Destination: 0.0.0.0/0 and Target as Internet Gateway. If missing, create route
 - aws ec2 create-route --route-table-id <route-table-id> --destination-cidr-block 0.0.0.0/0 --gateway-id <igw-id>
- Refresh browser page should display “Hello from your web server” append /mompopcafe/ to the end of the URL, you should see the café website.

```
{ec2-user@cli-host ~}$ aws ec2 describe-route-tables --filter "Name=association.subnet-id,Values=subnet-08e9c0fa8b4eb08"
{
  "RouteTables": [
    {
      "Associations": [
        {
          "SubnetId": "subnet-08e9c0fa8b4eb08",
          "AssociationState": {
            "State": "associated"
          },
          "RouteTableAssociationId": "rtbassoc-0877f6a36c98d8ed",
          "Main": false,
          "RouteTableId": "rtb-07b7e3d2aac8d7ecf"
        }
      ],
      "RouteTableId": "rtb-07b7e3d2aac8d7ecf",
      "VpcId": "vpc-8e402a13823b6cfb5",
      "PropagatingVgws": [],
      "Tags": [
        {
          "Value": "arn:aws:cloudformation:us-east-1:918678780405:stack/c193587a49732541135156891w918678780405/28cc43e8-12d7-11f1-b1e0-0eae087db67",
          "Key": "aws:cloudformation:stack-id"
        },
        {
          "Value": "VPC1 Public Route Table",
          "Key": "Name"
        },
        {
          "Value": "VPC1PublicRouteTable",
          "Key": "aws:cloudformation:logical-id"
        },
        {
          "Value": "c193587a49732541135156891w918678780405",
          "Key": "cloudlab"
        },
        {
          "Value": "c193587a49732541135156891w918678780405",
          "Key": "aws:cloudformation:stack-name"
        }
      ],
      "Routes": [
        {
          "GatewayId": "local",
          "DestinationCidrBlock": "10.0.0.0/16",
          "State": "active",
          "Origin": "CreateRouteTable"
        },
        {
          "GatewayId": "igw-04aaa51be128996f",
          "DestinationCidrBlock": "0.0.0.0/0",
          "State": "active",
          "Origin": "CreateRoute"
        }
      ],
      "OwnerId": "918678780405"
    }
  ]
}
```



- Troubleshooting Challenge 2: SSH still not working
 - i. Problem: Web access works now but SSH still times out
 - ii. Hint:
 - A. Re-establish the SSH connection to the CLI Host instance
 - B. Check the network ACL


```
aws ec2 describe-network-acls --filter "Name=association.subnet-id,Values='VPC1PublicSubnetID'" --query 'NetworkAcls[*].[NetworkAclId,Entries]'
```

look for deny rules blocking port 22
 - C. Remove Blocking Rule if found


```
aws ec2 delete-network-acl-entry --network-acl-id <acl-id> --rule-number <rule-number> --egress
```
 - D. Test SSH Again


```
ssh -i labsuser.pem ec2-user@<WebServerIP> -o ConnectTimeout=10
```
 - E. If successful, run hostname command it should indicate web-server as the hostname.

- Change the directory to the new directory
cd flowlogs
- List the S3 buckets
aws s3 ls
- Download the flow logs
aws s3 cp s3://<flowlog####>/ . -recursive
- Navigate to Log File Location
- cd <folder-name>
- ls
- repeat this until you reach
AWSLogs/<account-id>/vpcflowlogs/<region>/yyyy/mm/dd
- Extract the log files
gunzip *.gz

Step 2: View Log Structure

- Copy one log file name and run - head <filename>
- Search for rejected traffic – grep -rn REJECT
- Count Rejected Records - grep -rn REJECT . | wc -l
- Filter for SSH Port - grep -rn ' 22 ' . | grep REJECT
- Find your Public IP Address
- Log in to the AWS Management Console
- Go to EC2
- Choose Security Groups
- Select WebSecurityGroup
- Go to Inbound tab, choose to edit, then choose Add Rule
- For Source, select My IP
- Copy the populated IP address
- Filter Logs by your Public IP - grep -rn ' 22 ' . | grep REJECT | grep <ip-address>
- Verify Network Interface ID
aws ec2 describe-network-interfaces --filters "Name=association.public-ip,Values='<WebServerIP>'" --query
'NetworkInterfaces[*].[NetworkInterfaceId,Association.PublicIp]'

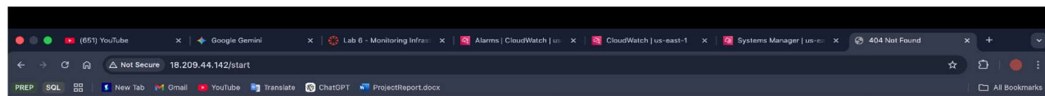
4. Given a cafe business application deployed on a web server configured to generate access and error logs, and the requirement for the cafe business client to monitor application and web server metrics, how would you design a monitoring and alerting system? Specifically, outline the steps you would take to ensure timely alerts to administrators when the application becomes unreachable to clients. Additionally, detail your approach to automatically backing up the EBS volume of the web server to take a snapshot in the event the web server transitions from a running state to a stopped state.
- Use AWS Systems Manager Run Command to install the CloudWatch Agent on Amazon EC2 instances
 - Monitor Application Logs using CloudWatch Agent and CloudWatch Logs
 - Monitor system metrics using CloudWatch Agent and CloudWatch Metrics
 - Create real-time notifications using CloudWatch Events
 - Track infrastructure compliance using AWS Config

Task 1: Install the CloudWatch Agent

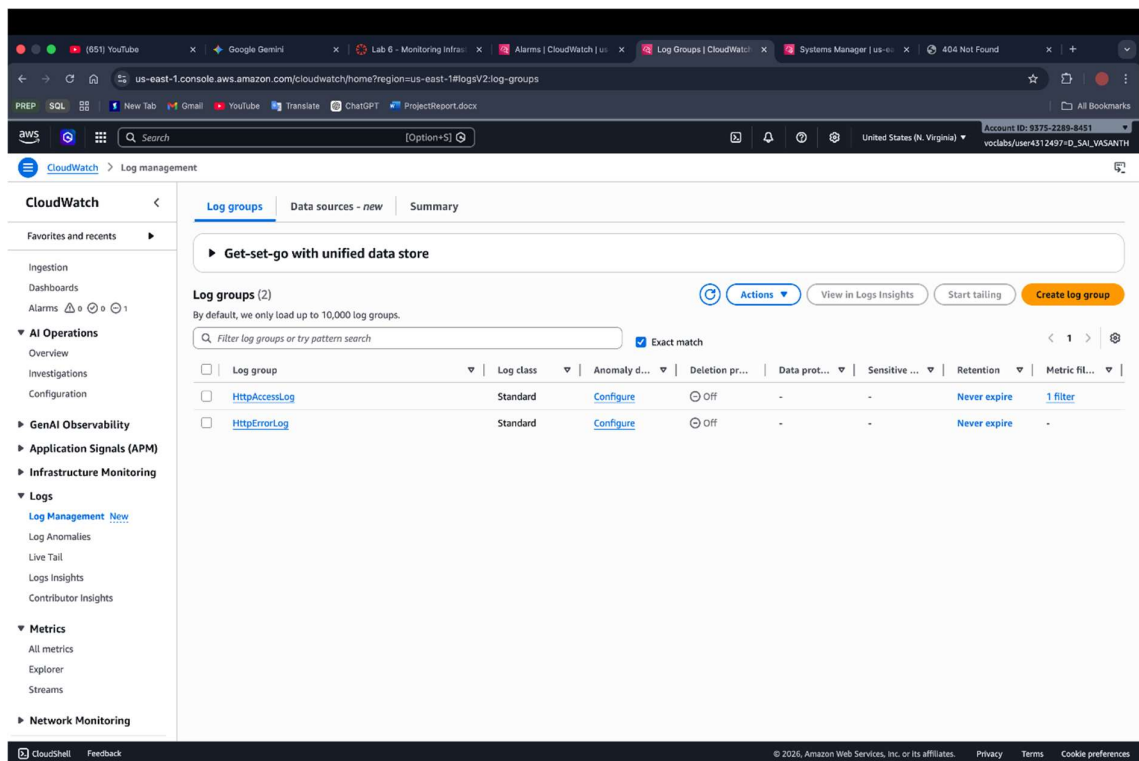
- The CloudWatch Agent can be used to collect metrics from Amazon EC2 instances and on-premises servers.
- Here we use AWS Systems Manager to install the CloudWatch Agent on an Amazon EC2 instance. You will configure it to collect both application and system metrics.

Task 2: Monitor Application Logs using CloudWatch Logs

- Amazon CloudWatch Logs to monitor applications and systems using log data. For example, CloudWatch Logs can track the number of errors that occur in your application logs and send you a notification whenever the rate of errors exceeds a threshold you specify.



- Open the CloudWatch console.
 - In the left navigation pane, expand Logs, choose Log groups.
- You should see two logs listed: HttpAccessLog and HttpErrorLog.



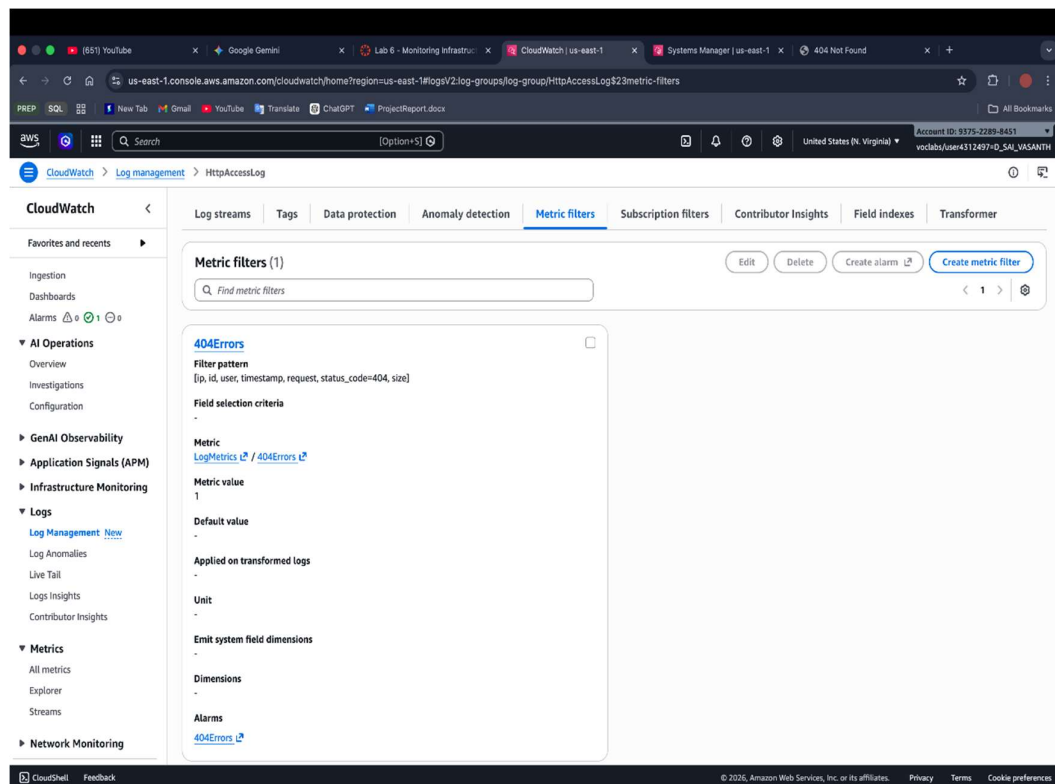
- Choose `HttpAccessLog`
- Choose the value (instance-id) displayed under **Logs Streams**.
- Log data should be displayed, consisting of **GET** requests that were sent to the web server.

Create a Metric Filter in CloudWatch Logs

- Filter to identify *404 Errors* in the log file. This would normally be an indication that the web server is generating invalid links that users are clicking.
- A filter pattern defines the fields in the log file and filters the data for specific values:

[ip, id, user, timestamp, request, status_code=404, size]

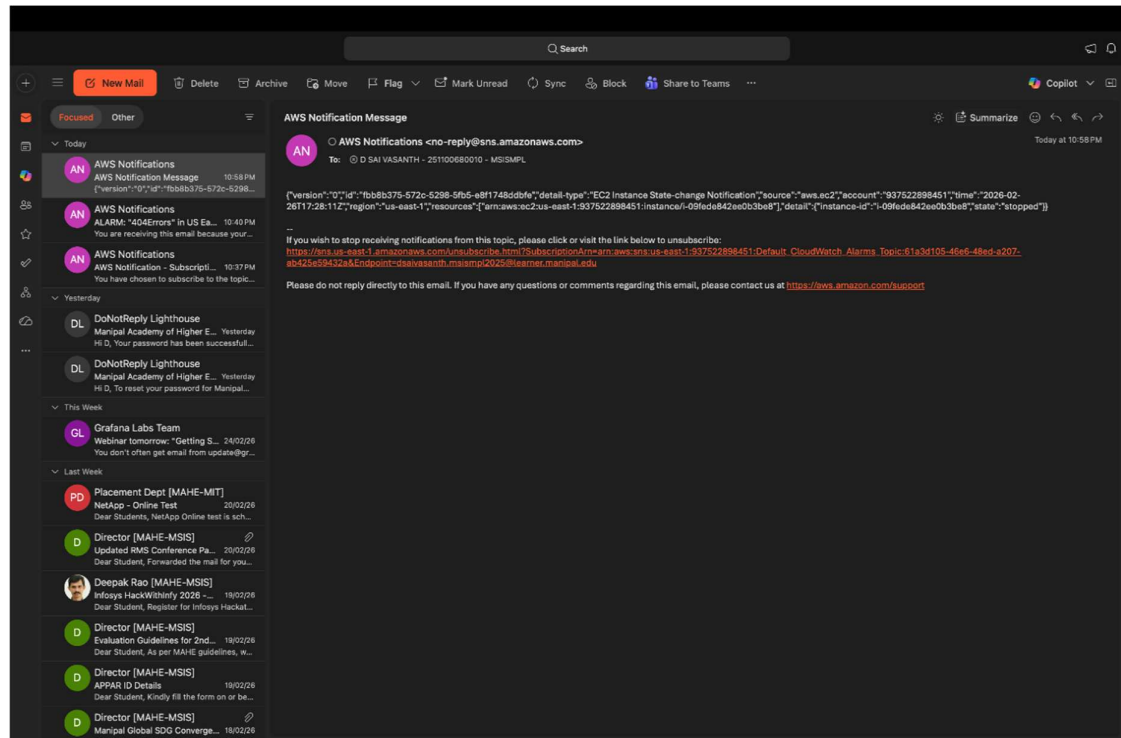
This tells CloudWatch Logs how to interpret the fields in the log data and defines a filter to only find lines with `status_code=404`, which indicates that a page was not found.



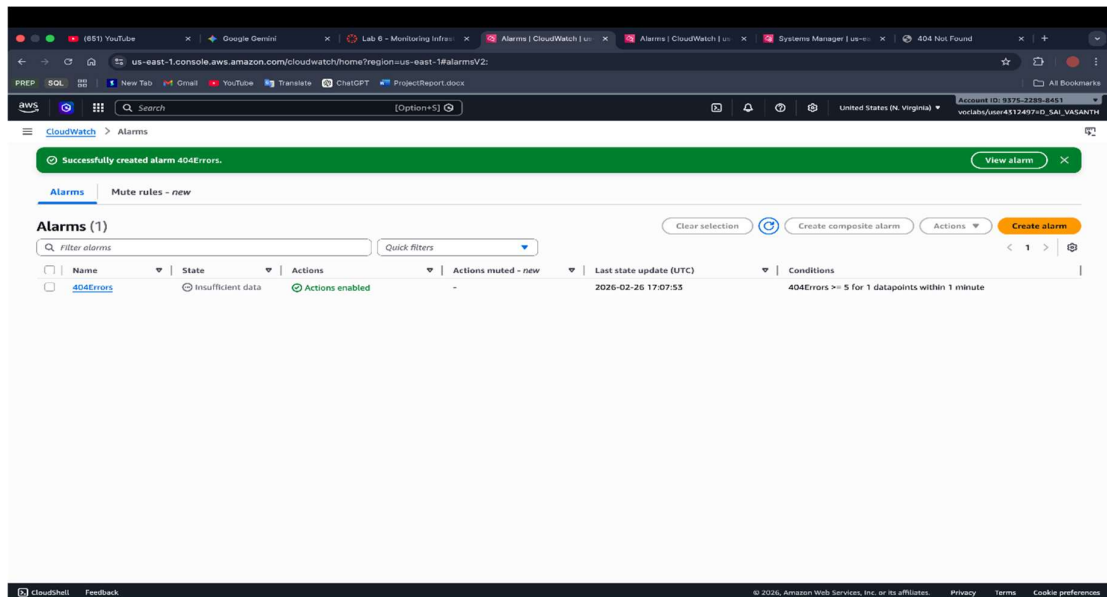
Create an Alarm using the Filter

You will now configure an Alarm to send a notification when too many 404 Not Found errors are received.

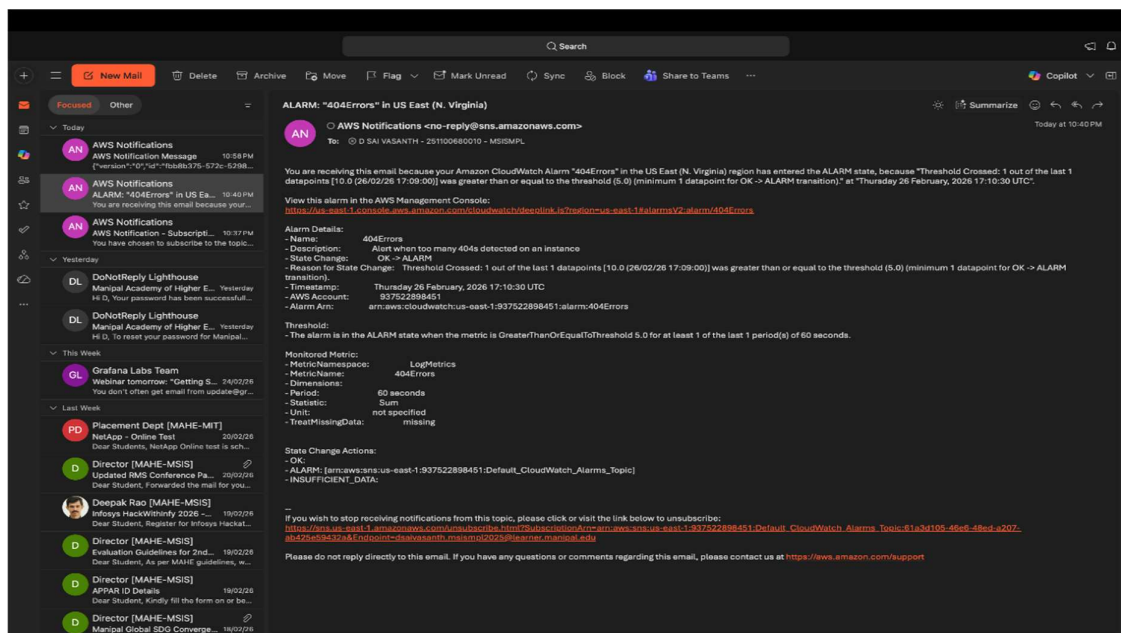
- Next to Metric filters, choose Create alarm configure
 - For Notification, configure Send a notification to the following SNS topic:
Create new topic
 - Email endpoints that will receive the notification: Enter an email address that you can access from the classroom
- Choose Create alarm
- Go to your email, look for a confirmation message and select the Confirm subscription link.



- Choose All Alarms



- You will now access the web server to generate log data.
- Return to the web browser tab with the web server.
- Wait 1-2 minutes for the Alarm to trigger. You can occasionally choose Refresh to update the status.
- The graph shown on the CloudWatch page should turn red to indicate that it is in the *ALARM* state.
- Check your email. You should have received an email titled *ALARM: "404 Errors"*.

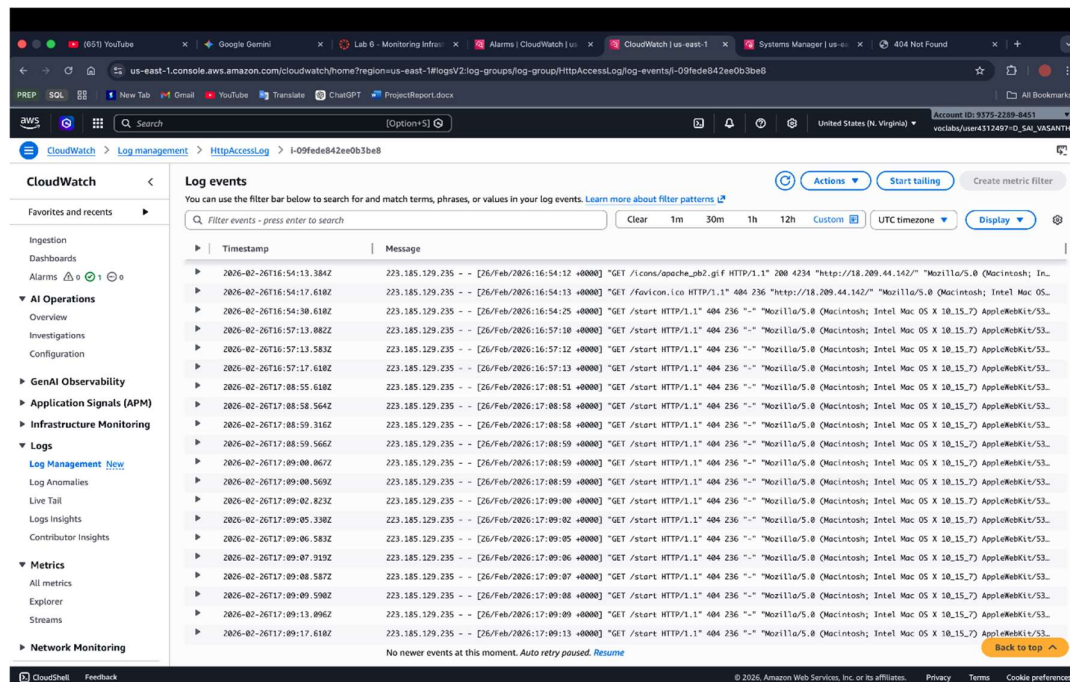


Task 3: Monitor Instance Metrics using CloudWatch

- Metrics are data about the performance of your systems. Amazon CloudWatch stores metrics for AWS services you use.
- select **CloudWatch** choose **All metrics**.

The lower half of the page will display the various metrics that have been collected by CloudWatch. Some are automatically generated by AWS while others were collected by the CloudWatch Agent.

- Choose **CWAgent**, then **device**, **fstype**, **host**, **path**.



Task 4: Create Real-Time Notifications

- Amazon CloudWatch Events delivers a near real-time stream of system events that describe changes in AWS resources.
- CloudWatch Events responds to these operational changes and takes corrective action as necessary, by sending messages to respond to the environment, activating functions, making changes, and capturing state information.
- In this task, you will create a real-time notification that informs you when an instance is Stopped or Terminated.

Review the Real-Time Notification Configuration

- In addition to receiving an email, you can configure Amazon Simple Notification Service (SNS) to send you a notification to your phone via SMS.

Task 5: Monitor for Infrastructure Compliance

- AWS Config is a service that enables you to assess, audit, and evaluate the configurations of your AWS resources.
- Config continuously monitors and records your AWS resource configurations and allows you to automate the evaluation of *recorded configurations* against *desired configurations*.

5. Suppose the cafe business application's web server is compromised by an unauthorized user within the account, who adds a new rule to the security group and updates images of the web application. The client expresses the need to audit actions performed on resources to identify vulnerabilities and details. How would you approach auditing these actions, and what steps would you propose to prevent similar unauthorized access in the future?
- Configure an AWS CloudTrail trail.
 - Analyze CloudTrail logs by using various methods to discover relevant information.
 - Import AWS CloudTrail log data into Amazon Athena.
 - Run queries in Amazon Athena to filter AWS CloudTrail log entries.
 - Resolve security concerns within the AWS account and on an EC2 Linux instance.

Task 1: Modify a security group and observe the website

- Choose the EC2 instance ie., Cafe Web Server instance and select the Security tab below and then choose WebSecurityGroup.

The screenshot shows the AWS Management Console interface for the security group **sg-027b65c43119484d5 - WebSecurityGroup**. The console displays the following details:

- Details:**
 - Security group name: WebSecurityGroup
 - Security group ID: sg-027b65c43119484d5
 - Description: Enable HTTP and SSH ingress
 - VPC ID: vpc-019be38f8869c59d3
 - Owner: 052458655550
 - Inbound rules count: 3 Permission entries
 - Outbound rules count: 1 Permission entry
- Inbound rules (3):**

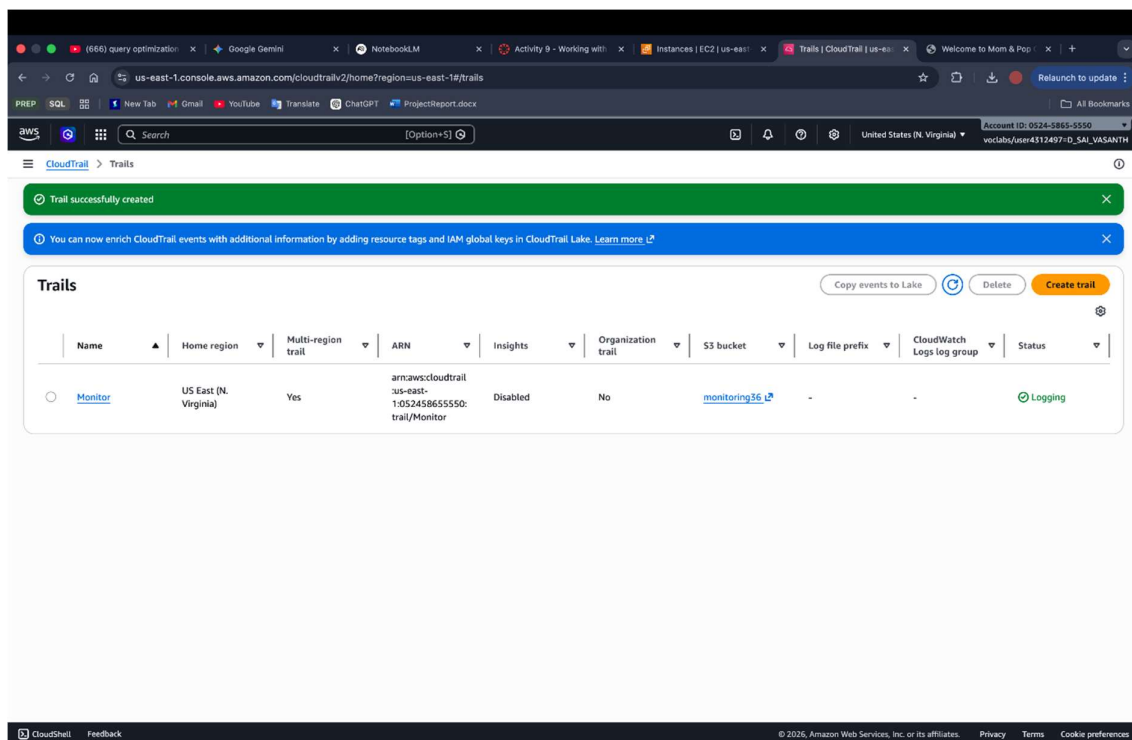
Name	Security group rule ID	IP version	Type	Protocol	Port range	Source
-	sgr-00c1060c0dd81f8dc	IPv4	SSH	TCP	22	0.0.0.0/0
-	sgr-0b8b6e5521d9a0d66	IPv4	HTTP	TCP	80	0.0.0.0/0
-	sgr-06b393478e9e1a8b2	IPv4	SSH	TCP	22	223.185.129.118/32

- Choose Edit inbound rules, then choose Add Rule and configure the rule as follows:
 - Port Range: 22
 - Source: My IP
- Observe the Café website: <http://<WebServerIP>/mompopcafe/>

Task 2: Create an AWS CloudTrail log and observe the hacked website

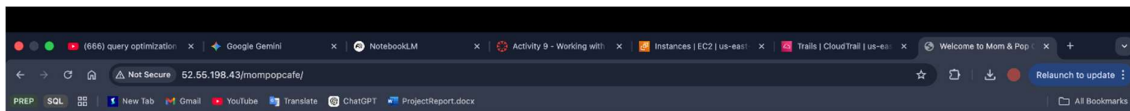
Create an AWS CloudTrail log:

- Configure the trail **Choose trail attributes**, set the **Trial name** as *Monitor*
- Storage location: Create a new S3 bucket
- Create trail.



Observe the hacked website:

- Return to the browser tab where you have the Café website open and refresh the page.



Task 3: Analyse the CloudTrail Logs by using grep

Connect to the Cafe Web Server host EC2 instance by using SSH:

- `chmod 400 labsuser.pem`
- `ssh -I labsuser.pem ec2-user@<public-ip>`

```
[ec2-user@web-server ~]$ mkdir ctraillogs
[ec2-user@web-server ~]$ cd ctraillogs
[ec2-user@web-server ctraillogs]$ aws s3 ls
2026-03-02 08:22:24 mompopangelles73289
2026-03-02 08:28:24 monitoring36
[ec2-user@web-server ctraillogs]$ aws s3 cp s3://monitoring###/ . --recursive
-bash: monitoring###: No such file or directory
[ec2-user@web-server ctraillogs]$ aws s3 cp s3://monitoring36/ . --recursive
download: s3://monitoring36/AWSLogs/052458655550/CloudTrail/us-east-1/2026/03/02/052458655550_CloudTrail/us-east-1/2026/03/02/052458655550_CloudTrail-us-east-1_20260302T0840Z_inZAdmAAo3KM.json.gz to AWSLogs/052458655550/CloudTrail/us-east-1/2026/03/02/052458655550_CloudTrail-us-east-1_20260302T0840Z_inZAdmAAo3KM.json.gz
download: s3://monitoring36/AWSLogs/052458655550/CloudTrail/eu-west-2/2026/03/02/052458655550_CloudTrail/eu-west-2/20260302T0835Z_38KOEe08Saiky3m.json.gz to AWSLogs/052458655550/CloudTrail/eu-west-2/2026/03/02/052458655550_CloudTrail-eu-west-2_20260302T0835Z_38KOEe08Saiky3m.json.gz
[ec2-user@web-server ctraillogs]$ ls
AWSLogs
[ec2-user@web-server ctraillogs]$ cd AWSLogs/
[ec2-user@web-server AWSLogs]$ ls
052458655550
[ec2-user@web-server AWSLogs]$ cd 052458655550/CloudTrail/us-east-1/2026/03/02/
[ec2-user@web-server 02]$ ls
052458655550_CloudTrail-us-east-1_20260302T0840Z_inZAdmAAo3KM.json.gz
[ec2-user@web-server 02]$ gunzip *.gz
[ec2-user@web-server 02]$ ls
052458655550_CloudTrail-us-east-1_20260302T0840Z_inZAdmAAo3KM.json
[ec2-user@web-server 02]$
```

Analyze the logs by using grep:

Analyze the logs by using AWS CLI cloudtrail commands

- It can be difficult to find specific information within a very large dataset.
- Athena is an interactive query service that makes it easy to analyse data in Amazon Simple Storage Service (Amazon S3) by using standard SQL.

- It can be difficult to find specific information within a very large dataset.
- Athena is an interactive query service that makes it easy to analyse data in Amazon Simple Storage Service (Amazon S3) by using standard SQL.

[illegible]

- The advantage of using Athena is that you can now run SQL queries over your log data.

- The advantage of using Athena is that you can now run SQL queries over your log data.

The screenshot shows the Amazon Athena Query Editor interface. On the left, the 'Data' sidebar displays the 'cloudtrail_logs_monitoring36' table with its schema. The main area shows a SQL query: `SELECT useridentity.userName, eventtime, eventsource, eventname, requestparameters FROM cloudtrail_logs_monitoring36 LIMIT 30`. Below the query, the 'Query results' tab is active, showing a table with 30 rows of data. The table has columns: #, userName, eventtime, eventsource, eventname, and requestparameters. The first four rows are visible, showing events from 's3.amazonaws.com' and 'ec2.amazonaws.com'.

#	userName	eventtime	eventsource	eventname	requestparameters
1		2026-03-02T08:45:16Z	s3.amazonaws.com	GetBucketAcl	{\"bucketName\":\"monitoring36\",\"Host\":\"monitoring36.s3.us-east-1.amazonaws.com\", \"act\":\"\"}
2		2026-03-02T08:49:29Z	logs.amazonaws.com	DescribeMetricFilters	{\"limit\":50}
3	chaos	2026-03-02T08:32:39Z	ec2.amazonaws.com	DescribeSecurityGroups	{\"securityGroupIdSet\":[],\"securityGroupIdSet\":[],\"filterSet\":{\"items\":{\"name\":\"ip-permission.cidr\",\"valueSet\":{\"items\":{\"value\":\"0.0.0.0/0\"}}}}}
4	chaos	2026-03-02T08:32:50Z	ec2.amazonaws.com	DescribeSecurityGroups	{\"securityGroupIdSet\":[],\"securityGroupIdSet\":[],\"filterSet\":{\"items\":{\"name\":\"ip-permission.cidr\",\"valueSet\":{\"items\":{\"value\":\"0.0.0.0/0\"}}}}}

Task 5: Further Analysis of the Hack and Improving Security

- Secure both your AWS account, and the web server instance.
- Check the OS users & Update SSH security.

The screenshot shows the Amazon Athena Query Editor interface with a large table of query results. The table has columns: userName, eventtime, eventsource, eventname, and requestparameters. The results show a series of events, including 'DescribeSecurityGroups' and 'DescribeProjects', with request parameters indicating IP permissions and security group IDs.

userName	eventtime	eventsource	eventname	requestparameters
	2026-03-02T08:45:16Z	s3.amazonaws.com	GetBucketAcl	{\"bucketName\":\"monitoring36\",\"Host\":\"monitoring36.s3.us-east-1.amazonaws.com\", \"act\":\"\"}
	2026-03-02T08:49:29Z	logs.amazonaws.com	DescribeMetricFilters	{\"limit\":50}
chaos	2026-03-02T08:32:39Z	ec2.amazonaws.com	DescribeSecurityGroups	{\"securityGroupIdSet\":[],\"securityGroupIdSet\":[],\"filterSet\":{\"items\":{\"name\":\"ip-permission.cidr\",\"valueSet\":{\"items\":{\"value\":\"0.0.0.0/0\"}}}}}
chaos	2026-03-02T08:32:50Z	ec2.amazonaws.com	DescribeSecurityGroups	{\"securityGroupIdSet\":[],\"securityGroupIdSet\":[],\"filterSet\":{\"items\":{\"name\":\"ip-permission.cidr\",\"valueSet\":{\"items\":{\"value\":\"0.0.0.0/0\"}}}}}
chaos	2026-03-02T08:33:01Z	ec2.amazonaws.com	DescribeSecurityGroups	{\"securityGroupIdSet\":[],\"securityGroupIdSet\":[],\"filterSet\":{\"items\":{\"name\":\"ip-permission.cidr\",\"valueSet\":{\"items\":{\"value\":\"0.0.0.0/0\"}}}}}
chaos	2026-03-02T08:33:13Z	ec2.amazonaws.com	DescribeSecurityGroups	{\"securityGroupIdSet\":[],\"securityGroupIdSet\":[],\"filterSet\":{\"items\":{\"name\":\"ip-permission.cidr\",\"valueSet\":{\"items\":{\"value\":\"0.0.0.0/0\"}}}}}
chaos	2026-03-02T08:33:24Z	ec2.amazonaws.com	DescribeSecurityGroups	{\"securityGroupIdSet\":[],\"securityGroupIdSet\":[],\"filterSet\":{\"items\":{\"name\":\"ip-permission.cidr\",\"valueSet\":{\"items\":{\"value\":\"0.0.0.0/0\"}}}}}
chaos	2026-03-02T08:39:51Z	ec2.amazonaws.com	DescribeSecurityGroups	{\"securityGroupIdSet\":[],\"securityGroupIdSet\":[],\"filterSet\":{\"items\":{\"name\":\"ip-permission.cidr\",\"valueSet\":{\"items\":{\"value\":\"0.0.0.0/0\"}}}}}
chaos	2026-03-02T08:40:03Z	ec2.amazonaws.com	DescribeSecurityGroups	{\"securityGroupIdSet\":[],\"securityGroupIdSet\":[],\"filterSet\":{\"items\":{\"name\":\"ip-permission.cidr\",\"valueSet\":{\"items\":{\"value\":\"0.0.0.0/0\"}}}}}
chaos	2026-03-02T08:40:14Z	ec2.amazonaws.com	DescribeSecurityGroups	{\"securityGroupIdSet\":[],\"securityGroupIdSet\":[],\"filterSet\":{\"items\":{\"name\":\"ip-permission.cidr\",\"valueSet\":{\"items\":{\"value\":\"0.0.0.0/0\"}}}}}
chaos	2026-03-02T08:40:26Z	ec2.amazonaws.com	DescribeSecurityGroups	{\"securityGroupIdSet\":[],\"securityGroupIdSet\":[],\"filterSet\":{\"items\":{\"name\":\"ip-permission.cidr\",\"valueSet\":{\"items\":{\"value\":\"0.0.0.0/0\"}}}}}
chaos	2026-03-02T08:40:37Z	ec2.amazonaws.com	DescribeSecurityGroups	{\"securityGroupIdSet\":[],\"securityGroupIdSet\":[],\"filterSet\":{\"items\":{\"name\":\"ip-permission.cidr\",\"valueSet\":{\"items\":{\"value\":\"0.0.0.0/0\"}}}}}
	2026-03-02T08:37:38Z	rekognition.amazonaws.com	DescribeProjects	
	2026-03-02T08:38:39Z	s3.amazonaws.com	GetBucketAcl	{\"bucketName\":\"monitoring36\",\"Host\":\"monitoring36.s3.us-east-1.amazonaws.com\", \"act\":\"\"}
chaos	2026-03-02T08:33:35Z	ec2.amazonaws.com	DescribeSecurityGroups	{\"securityGroupIdSet\":[],\"securityGroupIdSet\":[],\"filterSet\":{\"items\":{\"name\":\"ip-permission.cidr\",\"valueSet\":{\"items\":{\"value\":\"0.0.0.0/0\"}}}}}
chaos	2026-03-02T08:33:47Z	ec2.amazonaws.com	DescribeSecurityGroups	{\"securityGroupIdSet\":[],\"securityGroupIdSet\":[],\"filterSet\":{\"items\":{\"name\":\"ip-permission.cidr\",\"valueSet\":{\"items\":{\"value\":\"0.0.0.0/0\"}}}}}
chaos	2026-03-02T08:33:58Z	ec2.amazonaws.com	DescribeSecurityGroups	{\"securityGroupIdSet\":[],\"securityGroupIdSet\":[],\"filterSet\":{\"items\":{\"name\":\"ip-permission.cidr\",\"valueSet\":{\"items\":{\"value\":\"0.0.0.0/0\"}}}}}
chaos	2026-03-02T08:34:10Z	ec2.amazonaws.com	DescribeSecurityGroups	{\"securityGroupIdSet\":[],\"securityGroupIdSet\":[],\"filterSet\":{\"items\":{\"name\":\"ip-permission.cidr\",\"valueSet\":{\"items\":{\"value\":\"0.0.0.0/0\"}}}}}
chaos	2026-03-02T08:34:21Z	ec2.amazonaws.com	DescribeSecurityGroups	{\"securityGroupIdSet\":[],\"securityGroupIdSet\":[],\"filterSet\":{\"items\":{\"name\":\"ip-permission.cidr\",\"valueSet\":{\"items\":{\"value\":\"0.0.0.0/0\"}}}}}
chaos	2026-03-02T08:34:33Z	ec2.amazonaws.com	DescribeSecurityGroups	{\"securityGroupIdSet\":[],\"securityGroupIdSet\":[],\"filterSet\":{\"items\":{\"name\":\"ip-permission.cidr\",\"valueSet\":{\"items\":{\"value\":\"0.0.0.0/0\"}}}}}

6. The cafe business client wishes to update their website with new recipes, images, and videos. They have hired an external media company to create these images and videos. The client requires a secure file sharing mechanism to transfer data from the external company to the business owner. Additionally, they want to be notified immediately when images are added, updated, or deleted. As a cloud operations engineer, propose a solution to meet the business's needs.

- Use the *s3api* and *s3* AWS CLI commands to create and configure an Amazon S3 bucket.
- Configure an Amazon S3 bucket for file sharing with an external user.
- Secure an Amazon S3 bucket for different access requirements by using Amazon S3 permissions.
- Configure event notification on an Amazon S3 bucket.

Task 1: Connect to the AWS CLI Host instance by using SSH

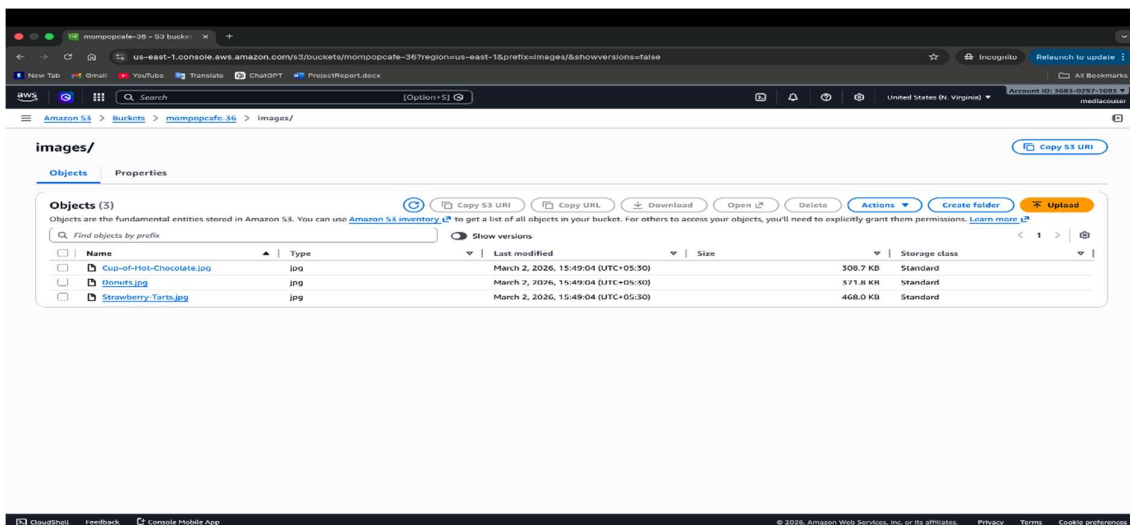
Configure the AWS CLI on the CLI Host EC2 Instance

```
[ec2-user@ip-10-200-0-102 ~]$ aws configure
AWS Access Key ID [AKIAVQLERLDKXVHGDRPH]:
AWS Secret Access Key [*****]:
Default region name [us-east-1]:
Default output format [json]:
[ec2-user@ip-10-200-0-102 ~]$ cat ~/.aws/credentials
[default]
aws_access_key_id = AKIAVQLERLDKXVHGDRPH
aws_secret_access_key = IfmWq2CuKqg1bNb1a9WtbnY0DQwT8lP+nT
[ec2-user@ip-10-200-0-102 ~]$ cat ~/.aws/config
[default]
aws_access_key_id = AKIAVQLERLDKXVHGDRPH
aws_secret_access_key = IfmWq2CuKqg1bNb1a9WtbnY0DQwT8lP+nT
[ec2-user@ip-10-200-0-102 ~]$ aws sts get-caller-identity
{
  "Account": "366382971893",
  "User": "arn:aws:iam::366382971893:user/mediscouser",
  "Arn": "arn:aws:iam::366382971893:user/mediscouser"
}
[ec2-user@ip-10-200-0-102 ~]$ cd new-images/
[ec2-user@ip-10-200-0-102 new-images]$ aws s3api put-object --bucket monppocafe-36 --key images/Caramel-Delight.jpg --body ~/new-images/Caramel-Delight.jpg
{"ETag": "\"31ac380ea1924a88c978f186eef3d77\"", "ServerSideEncryption": "AES256"}
[ec2-user@ip-10-200-0-102 new-images]$
```

Task 2: Create and initialize the Amazon S3 share bucket

```
[ec2-user@ip-10-200-0-102 ~]$ nano ~/.aws/credentials
[ec2-user@ip-10-200-0-102 ~]$ cat ~/.aws/credentials
[default]
aws_access_key_id = AKIAVLQERLDK3NWK3N1N7N
aws_secret_access_key = IfxMvQ3CUpAQ18Nbia9HvibnzY00QwrIR1PnT
[ec2-user@ip-10-200-0-102 ~]$ aws sts get-caller-identity
{
  "Account": "368302971093",
  "UserId": "AIDAVLQERLDK3NWK3N1N7N",
  "Arn": "arn:aws:iam::368302971093:user/mediacouser"
}
[ec2-user@ip-10-200-0-102 ~]$ cd new-images/
[ec2-user@ip-10-200-0-102 new-images]$ aws s3api put-object --bucket mompopcafe-36 --key images/Caramel-Delight.jpg --body ~/new-images/Caramel-Delight.jpg
{
  "ETag": "\"31ac38da619244b8ce786f186e4f3df7\"",
  "ServerSideEncryption": "AES256"
}
[ec2-user@ip-10-200-0-102 new-images]$ aws s3api get-object --bucket mompopcafe-36 --key images/Donuts.jpg Donuts.jpg
{
  "AcceptRanges": "bytes",
  "ContentType": "image/jpeg",
  "LastModified": "Mon, 02 Mar 2026 10:19:04 GMT",
  "ContentLength": 308753,
  "ETag": "\"408b0bcc53cb5ab713c967dc1422b4fa\"",
  "ServerSideEncryption": "AES256",
  "Metadata": {}
}
[ec2-user@ip-10-200-0-102 new-images]$
```

- Use the AWS `s3` CLI to create the Amazon S3 share bucket and initialize it with some images
- Then list the bucket contents to verify the success of your actions.



Task 3: Review the media company user and permissions

- The user provides a way for the media company to use the AWS Management Console or the AWS CLI to upload and modify images in the S3 share bucket.
- Review the permissions that the *mompopuser* and *mediacouser* users have been granted to access the S3 bucket.

Review the mompopuser IAM User & mediaco IAM Group:

Permissions are defined by policies attached to the user directly or through groups.

Filter by Type: All types

Policy name | Type | Attached via

- IAMUserChangePassword** (AWS managed) | Group: [mediaco](#)

```

1- {
2-   "Version": "2012-10-17",
3-   "Statement": [
4-     {
5-       "Effect": "Allow",
6-       "Action": [
7-         "iam:ChangePassword"
8-       ],
9-       "Resource": [
10-        "arn:aws:iam:*:user/${aws:username}",
11-        "arn:aws:iam:*:user/*/${aws:username}"
12-      ]
13-     },
14-     {
15-       "Effect": "Allow",
16-       "Action": [
17-         "iam:GetAccountPasswordPolicy"
18-       ],
19-       "Resource": "*"
20-     }
21-   ]
22- }

```
- mediaCoPolicy** (Customer inline) | Group: [mediaco](#)

▶ Permissions boundary (not set)

▼ Generate policy based on CloudTrail events

You can generate a new policy based on the access activity for this user, then customize, create, and attach it to this role. AWS uses your CloudTrail events to identify the services and actions used and generate a policy. [Learn more](#)

Review the mediacouser IAM User:

Summary

ARN: [arn:aws:iam::368302971093:user/mediacouser](#)

Created: March 02, 2026, 15:40 (UTC+05:30)

Console access: Enabled without MFA

Last console sign-in: Never

Access key 1: AKIAVLQERLDKYHIGDRPH - Active
 Ⓛ Never used. Created today.

Access key 2: Create access key

Permissions | Groups (1) | Tags (2) | **Security credentials** | Last Accessed

Console sign-in [Manage console access](#)

Console sign-in link: <https://368302971093.signin.aws.amazon.com/console>

Console password: Updated 25 minutes ago (2026-03-02 15:41 GMT+5:30)

Last console sign-in: Ⓛ Never

Multi-factor authentication (MFA) (0)

Use MFA to increase the security of your AWS environment. Signing in with MFA requires an authentication code from an MFA device. Each user can have a maximum of 8 MFA devices assigned. [Learn more](#)

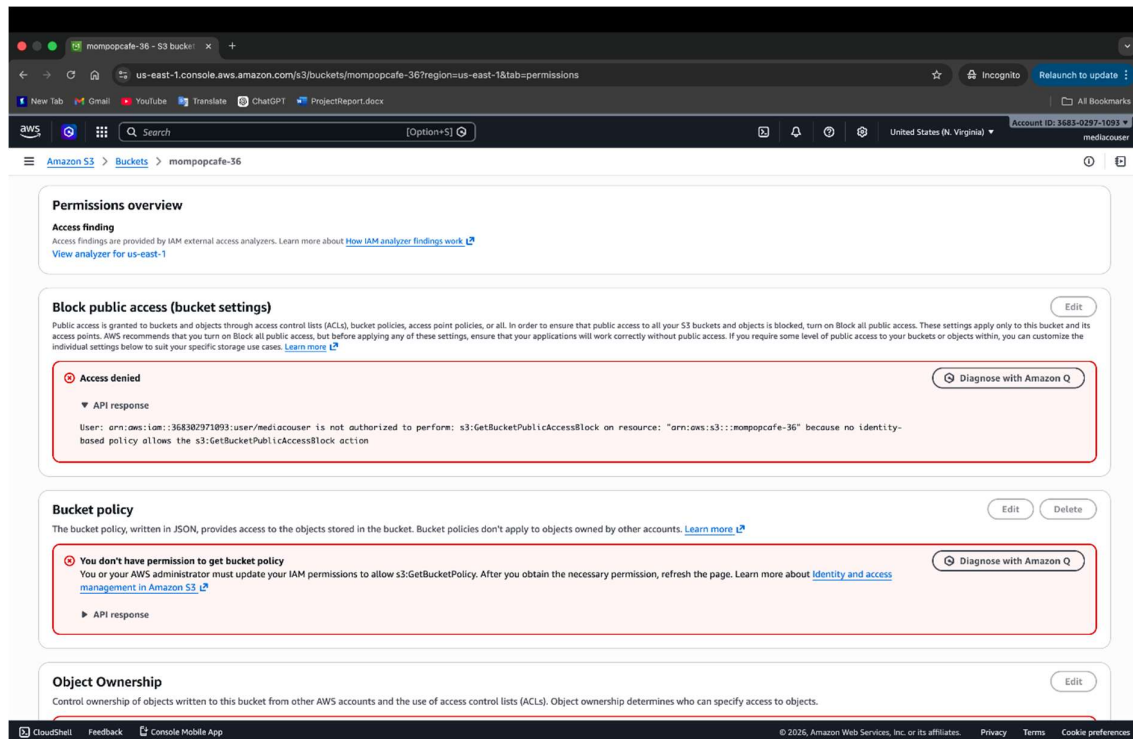
Type	Identifier	Certifications	Created on
No MFA devices. Assign an MFA device to improve the security of your AWS environment.			

[Assign MFA device](#)

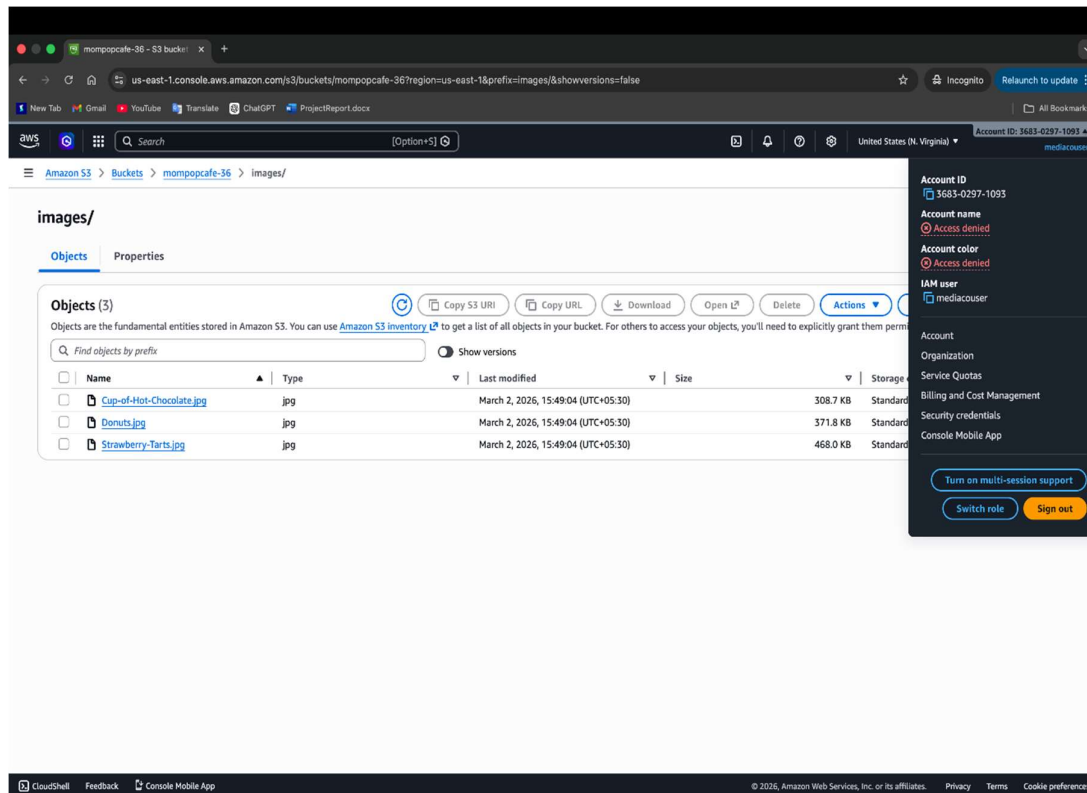
Access keys (1) [Create access key](#)

Use access keys to send programmatic calls to AWS from the AWS CLI, AWS Tools for PowerShell, AWS SDKs, or direct AWS API calls. You can have a maximum of two access keys (active or inactive) at a time. [Learn more](#)

Test the mediacouser permissions:



- In the browser tab in incognito, paste the URL, browse to <https://aws.amazon.com/console/>
- In the Sign in page, enter the following details, choose IAM user.
 - For Account ID or alias, Account ID
 - For IAM username, enter mediacouser
 - For Password, enter Training1!
 - Choose Sign in
- Goto S3, choose the mompopcafe-xxxxnnn link, where xxxnnn corresponds to your bucket's unique name.
- Choose the images link.
- See the list of images that were uploaded when you initialized the bucket in Task 2.
- Test the view use case. Choose and open Donuts.jpg.

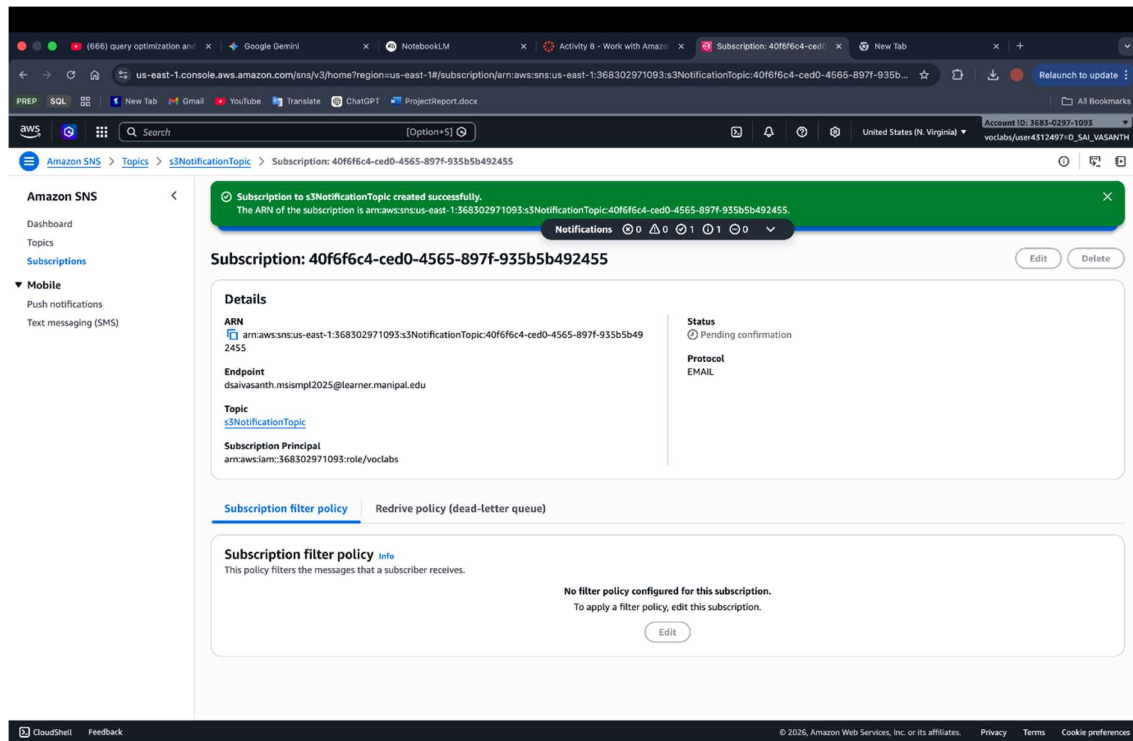


Task 4: Configure event notifications on the Amazon S3 share bucket

- Configure the Amazon S3 share bucket to generate an event notification to an Amazon SNS topic whenever the bucket's contents change.
- The topic then sends an email message to its subscribed users with the notification message.

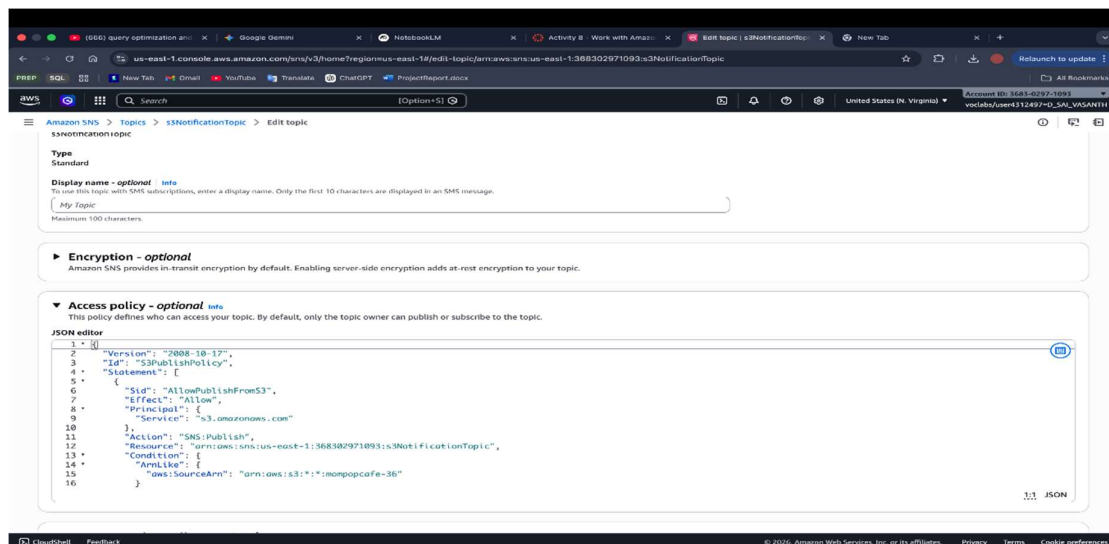
Create and configure the s3NotificationTopic:

- Subscribe Pop to the topic as the mompopuser who will receive the event notifications from the S3 share bucket.
- Choose **Create subscription**, the **topic ARN** box, the **s3NotificationTopic** already appears.
- In the **Protocol** menu, select **Email**, enter an email address that you can access.
- Open the email message and choose Confirm subscription.

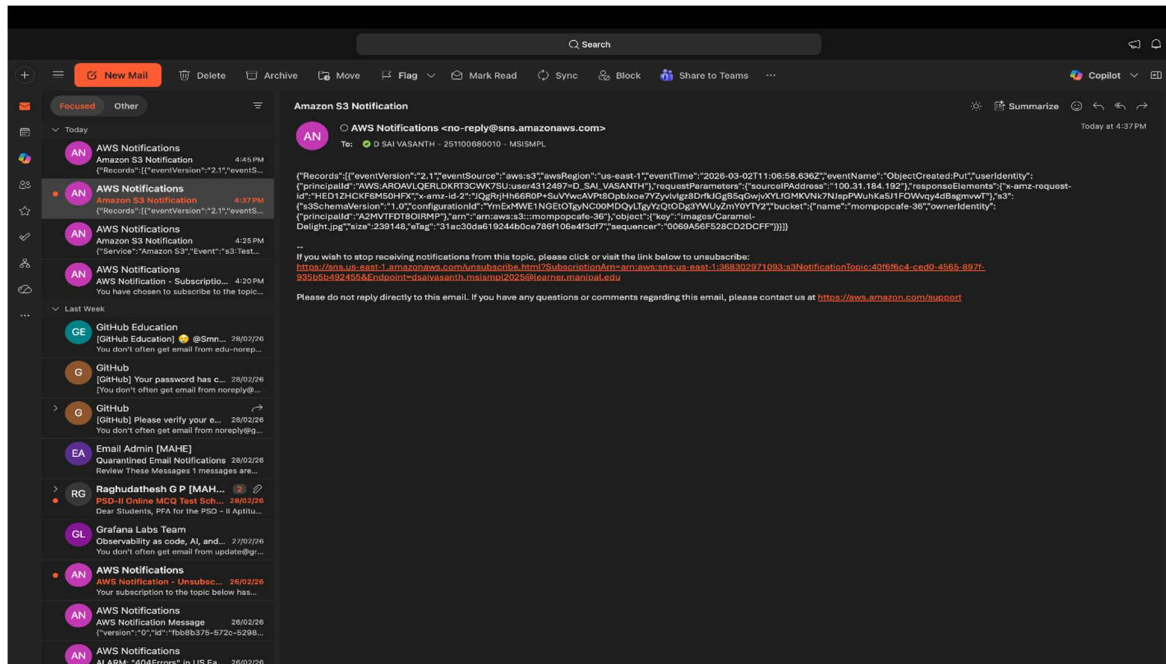


Add an event notification configuration to the S3 bucket:

- Create an event notification configuration file that identifies the events that Amazon S3 will publish and the topic destination where Amazon S3 will send the event notifications.
- Then use the *s3api* CLI to associate this configuration file with the Amazon S3 share bucket.



- Associate the event configuration file with the S3 share bucket.
- Wait a few moments and then check the inbox for the email address that you used to subscribe to the topic.
- See an email message with the subject *Amazon S3 Notification*.



Task 5: Test the Amazon S3 share bucket event notifications

- Test the configuration of the S3 share bucket event notification by performing the use cases that the *mediacouser* expects to perform on the bucket.
- These actions include putting and deleting objects in the bucket, which should generate email notifications to Pop.